

# POCKET SMART AI — Your Smart Budget & Recommendation Assistant

## Phase 1: Brainstorming & Ideation

St. George's Arts and Science College (UNM1727) · TN Skills | Generative AI · Team ID: SWTID-2026-2087

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### 1.1 Purpose of This Phase

Before any design or code work began, the team held structured brainstorming sessions to explore the problem space, generate candidate ideas, and converge on a concept that was both useful and achievable within the project timeline. This document records the ideation trail that led to Pocket Smart AI.

### 1.2 Problem Space Exploration

The team started from a broad observation: most budgeting tools are historical — they record what a user has already spent. The group identified a gap in **forward-looking** budgeting, where a user knows a fixed total amount up front and must decide how to split it across several related purchase needs before any money is spent.

- How do people currently plan a fixed budget across multiple items (e.g. furnishing a room, organizing an event)?
- Why do generic expense trackers fall short for this kind of planning?
- Could an AI-assisted allocation and recommendation layer make this decision easier and faster?
- What would make the output trustworthy and easy to act on for a non-technical user?

### 1.3 Idea Generation

The team generated and discussed multiple candidate directions before settling on the final concept:

| Idea                                | Description                                                                           | Decision                                             |
|-------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------|
| Simple expense tracker              | Log income/expenses after they occur, show charts.                                    | Rejected — too common, no planning value.            |
| Fixed-budget multi-category planner | User enters one total budget plus item requirements; system allocates and recommends. | Selected — addresses a real gap.                     |
| Pure chatbot advisor                | Free-form chat only, no structured output.                                            | Rejected — output would be hard to act on.           |
| Investment-only planner             | Focused narrowly on gold/jewelry investment.                                          | Merged as one planner domain, not the whole product. |

### 1.4 Concept Selected: Pocket Smart AI

The chosen concept, **Pocket Smart AI**, is a web-based budgeting and recommendation assistant. Instead of limiting the user to recording expenses after the fact, it focuses on planning: the user provides a budget and purchasing needs, and the application organizes the available amount across relevant categories and returns structured, forward-looking purchasing guidance.

### 1.5 Target Users Identified

- Students and individual consumers managing a personal budget.
- Households planning home-decor or furnishing purchases.
- Event planners and students organizing an event on a fixed budget.

- Anyone who needs to distribute one total amount across several connected purchase requirements.

## 1.6 Brainstormed Use-Case Scenarios

### Scenario 1 — End-user purchase planning

A user has a fixed amount and several items to purchase. They enter the overall budget and requirements, and the system presents an organized allocation and recommendation view.

### Scenario 2 — Home-decor planning

A user plans a room or home purchase across categories such as furniture, lighting, and accessories, structuring the available budget before purchasing.

### Scenario 3 — Event planning

An event planner or student distributes a fixed amount among categories such as catering, venue, and entertainment before any commitments are made.

### Scenario 4 — Educational usage

The application can be demonstrated in a classroom or project lab to teach requirements analysis, UI design, input validation, workflow design, budgeting logic, and deployment practice.

## 1.7 Key Capabilities Brainstormed

- A simple user access flow (registration and login).
- Structured input of a total budget and category-specific requirements.
- AI-assisted allocation logic that splits the budget across categories.
- Recommendation-oriented output rather than raw numbers alone.
- A dashboard to review and revisit past plans.
- Support for multiple planning domains: Home Interior, Party, and Jewelry.

## 1.8 Feasibility & Direction Check

The team assessed the idea against available skills (web development, basic backend logic, and prompt-based AI reasoning), the project timeline, and the availability of a generative-AI recommendation engine. The concept was judged achievable as a browser-based application with an AI reasoning layer handling allocation and recommendation text, so it was carried forward into formal requirement analysis.

## 1.9 Outcome of This Phase

A validated product concept — **Pocket Smart AI** — with an initial set of target users, use-case scenarios, and a working list of desired capabilities, ready to be converted into formal functional and non-functional requirements in the next phase.